Dhruv Doshi

work@doshidhruv.com
doshidhruv.com
+1-902-989-1274 | Toronto, Canada | Montreal / Global

Skills

- **Quant Research**: intraday signals, short-horizon alpha, CTA trend signals, mean reversion, momentum, event studies
- **Statistical Methods**: hypothesis testing, robustness checks, Monte Carlo simulation, bootstrapping, regression, anomaly detection
- **Market Data**: intraday bars, time-series processing, data quality checks, live-vs-sim comparison, performance attribution
- **Trading Systems**: backtesting, risk controls, position sizing, execution monitoring, live strategy dashboards, alerting
- **Programming**: Python, Go, SQL, Pandas, NumPy, SciPy, Scikit-learn, Jupyter, REST APIs, Linux, Git
- **Data Platforms**: Kafka, Airflow, Snowflake, PostgreSQL, Redis, Elasticsearch, ETL, schema design, streaming pipelines
- **Production Research**: CI/CD, reproducible experiments, logging, monitoring, SLOs, incident response, runbooks

Experience

- **Royal Bank of Canada (RBC)** — Toronto, Canada
  Staff Software Developer & Enterprise Architect – Data / Quant Systems — *Nov 2022 – Present*
  - **Large-Scale Time-Series Data Platform**: Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas for time-series analytics, anomaly detection, alerting, SLOs, and live-vs-baseline monitoring.
  - **Research-Style Workflow Platform**: Led 0-to-1 delivery of patent-pending blueprint platform (Node.js/TypeScript, React, REST APIs) with rules-based recommendation engine, real-time data rendering, reproducible outputs, and reusable components adopted across engineering teams.
  - **Robustness, Monitoring & Production Checks**: Built operational controls for production systems including data-readiness checks, traceability, logging, alerting, runbooks, regression-style validation, and escalation paths for critical components.
  - **Requirements Analysis & Technical Communication**: Work directly with architecture, infrastructure, product, security, and risk stakeholders to analyze requirements, present results, define next steps, and drive projects through full engineering lifecycle.
  - **Technical Leadership**: Led 5 engineers through sprint planning, code reviews, design discussions, testing expectations, ADRs, documentation, and cross-team standards adopted by 20+ teams.
- **HealthCard (Acquired Startup)** — Remote
  Blockchain Architect & First Full Stack Engineer — *Aug 2021 – Jul 2022*
  - **Statistical Trust-Score Workflows**: Designed validation and trust-score logic for real-time premium verification and fraud-resistant claims processing; joined as first engineer and built React/TypeScript frontend, Node.js APIs, AWS EKS deployment, and audit logging.
  - **ML/Data Pipeline Integration**: Integrated transformer-based document classification and monitoring for medical/insurance records, reducing manual review by 60%; built production APIs and feedback loops for credential verification.
- **CryptoVantage** — Netherlands (Remote)
  Blockchain Developer — *Jul 2020 – Dec 2020*
  - **Real-Time Risk Scoring**: Built on-chain analytics in Go/JavaScript with Ethereum event streaming, wallet clustering, and risk scoring for AML-style transaction monitoring and suspicious-pattern detection.
- **Dalhousie University** — Halifax, Canada
  Research & Teaching Assistant — *Jan 2021 – Aug 2022*
  - **Algorithms, Systems & Cloud**: Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

Projects

- **Automatic Trading Algorithm**: Sub-second latency personal project integrating live brokerage APIs and real-time intraday market data; momentum/mean-reversion signals with automated risk management, position sizing, and performance monitoring.
- **Portfolio Risk Optimization Framework**: Python research framework for market data ingestion, portfolio allocation, backtesting, Sharpe analysis, Monte Carlo simulation, risk-adjusted signals, and reproducible quantitative experiments.
- **Intraday Strategy Validation Toolkit**: Prototype for comparing simulated vs live strategy behavior using bar-level features, slippage assumptions, drawdown checks, rolling metrics, and statistical significance filters.

Education

- **Dalhousie University** — Halifax, Canada
  MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing — *Jan 2021 – Aug 2022*
- **University of Toronto** — Toronto, Canada
  Enterprise Architecture Certification (3-semester program) — *Completed 2024*
- **Gujarat Technological University** — Ahmedabad, India
  B.E. Computer Engineering, GPA: 3.92/4.0 — *Aug 2016 – Aug 2020*